# Polarization of Light

Perhaps you have seen a display of polarized sunglasses in a store. You can quickly test to see if the glasses are really polarized by looking through the lenses of two pairs of glasses and rotating one pair by 90° (π/2 radians). If both pairs of glasses are polarized, the lenses will appear to go black. Why is that?

To explain the darkened lenses, we need to think of the light as an electromagnetic wave. An electromagnetic wave has varying electric and magnetic fields perpendicular to the direction the wave is traveling. This experiment focuses only on the electric field variation, represented by a vector. Light emitted from a typical source such as a flashlight is randomly polarized, meaning that the electric field vector points in varying directions.

An ideal polarizing filter will remove all but the electric fields that are parallel to the axis of the filter. The light remaining is then said to be *polarized*. A second filter can be used to detect the polarization; in this case, the second filter is called an *analyzer*. The transmission through the second filter depends on the angle between its axis and the axis of the first filter. In this experiment you will study the relationship between the light intensity transmitted through two polarizing filters and the angle between the filter axes.

In the 1800's Malus proposed a law to predict light transmission through two polarizing filters. The relationship is

$$I = I_0 \cos^2 \theta$$

where $I_0$ is the intensity when the angle $\theta$ between the polarizer axes is zero. In this experiment, you will see if this law is useful in describing your polarizing filters.

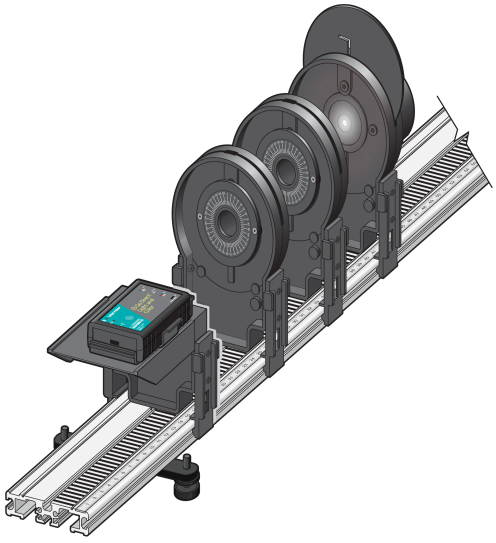


*Figure 1*

## OBJECTIVES

- Observe the change in light intensity of light passing through crossed polarizing filters.
- Measure the transmission of light through two polarizing filters as a function of the angle between their axes and compare it to Malus's law.

## MATERIALS

Chromebook, computer, **or** mobile device
Graphical Analysis 4 app
Go Direct Light and Color
Optics Expansion Kit: light source assembly and light sensor holder
Polarizer/Analyzer Set
Vernier Dynamics Track

## PRELIMINARY QUESTIONS

1. Place one Adjustable Analyzer from the Polarizer/Analyzer Set on top of a second Adjustable Analyzer so you can look through both of them. Place the pointer of the polarizer at 0° and rotate the pointer on the analyzer to 90°. What do you notice when the polarizer and analyzer are at right angles to one another?

2. Rotate the analyzer back to 0° and in alignment with the polarizer. Observe the transmitted light intensity as you rotate the analyzer by 180°. Make a qualitative graph of the transmitted light intensity you observed as a function of the angle between the polarizer and analyzer.

3. Based on the graph you just drew, construct a mathematical model for the transmission of light through the polarizer and analyzer as a function of the angle between the polarizer and analyzer. The model should minimally reproduce the observed bright and dark angles and be different from Malus's law.

## PROCEDURE

1. Set up the equipment.
   a. The location of the light source assembly can be read using the notched arrow on the base of the holder. Attach the light source assembly to the Dynamics Track at the 20 cm mark, directed along the track toward higher values. Set the disk to its open circle, to fully expose the LED. Connect power and turn on the LED. **Caution**: *The LED is bright*. **Note**: The brightness of the LED light source varies when it is first turned on. Make a note of the current time so that you can ensure 15 minutes have passed before you begin making the measurements.
   b. Set the pointer to 0° on one Adjustable Analyzer (i.e., the polarizer) and place it immediately adjacent to the light source, oriented so that the scale is visible.
   c. Place a second Adjustable Analyzer (i.e., the analyzer) further down the track. Orient it so the scale is visible. Light will pass through the polarizer and analyzer, with the amount of transmitted light depending on the relative angles of the polarizing sheets.
   d. Launch Graphical Analysis. Connect Go Direct Light and Color to your Chromebook, computer, or mobile device.

e. Attach the sensor cradle to the light sensor bracket, and place the light sensor in the cradle (see Figure 1). Position the light sensor assembly downstream of the analyzer.

   If you are not in a dark room, minimize the distance between equipment. This reduces the amount of stray light entering the light sensor, ensuring the sensor is only reading the intensity of the light source.

f. Rotate the analyzer and make a visual observation of how the light intensity varies with angle, reaching two maxima per full rotation.

2. Set up the data-collection mode.
   a. Click or tap Mode to open Data-Collection Settings. Change mode to Events Based.
   b. Enter **Angle** as the Event Name and **degree** as the Units. Click or tap Done.

3. Zero the light sensor.
   a. Keep the polarizer at 0° and rotate the analyzer to 90°. In this state, the polarizing axes are at right angles. Very little light should get through the pair.
   b. Define the light level as zero by clicking or tapping the Illumination meter and choosing Zero. The intensity reading should now be near zero.

4. You are now ready to collect data. In Events with Entry mode, data will only be recorded when you click or tap Keep.
   a. Return the analyzer to the parallel position, 0°, and click or tap Collect to start data collection.
   b. Click or tap Keep. Enter **0** for the angle click or tap Keep Point to store the data pair. Since this is the maximum illuminance, also record this value in your data table.

5. Rotate the analyzer by 15°, click or tap Keep, enter **15** for the angle and then click or tap Keep Point to store the data pair. Repeat this process, entering **30** for the next angle, and so forth, until you have rotated the analyzer through one revolution, or 360°. After the last point, click or tap Stop to stop data collection.

## DATA TABLE

| | |
|---|---|
| Maximum illuminance | |

## ANALYSIS

1. Describe your graph of illuminance *vs*. angle, giving important patterns and points.

2. In the 1800's, Étienne-Louis Malus proposed

$$I = I_0 \cos^2 \theta$$

to predict the light transmission through two polarizing filters, where $I_0$ is the intensity when the angle $\theta$ between the polarizer axes is zero.

You can use the Curve Fit feature of Graphical Analysis to superimpose this relationship on your experimental data.

   a. Click or tap Graph Tools, , and choose Apply Curve Fit.
   b. Select Cosine Squared as the curve fit. Click or tap Apply. The equation and coefficients are displayed on the screen. Graphical Analysis gives the parameters for the equation a * cos(b$x$+c)$^2$ + d. In this equation, $x$ represents the angle $\theta$ between the polarizers, a represents the maximum intensity $I_0$, and d represents the background illuminance.

3. Compare your data to the fit parameters.
   a. Compare the fit parameter a to the maximum illuminance that you recorded in your data table.
   b. Since you zeroed the light sensor before collecting data, the background light intensity is zero. Compare the value of parameter d to zero.
   c. Parameter b should be close to 1 and parameter c should be close to zero. How well do these parameters match up?

4. According to the curve fit on your graph, are your data consistent with Malus's law?

5. Now compare your data to the model you developed in Preliminary Question 3. Do this in any way you or your instructor chooses. Does your model match the data qualitatively?

## EXTENSIONS

1. Polarized sunglasses selectively remove glare caused by light reflected by horizontal surfaces, such as the hood of a car or a wet highway. For polarized sunglasses to be effective, the glare must be polarized. Devise an experiment to measure how strongly polarized the glare is compared to sunlight or to a flashlight. You will need a smooth horizontal surface to create glare, a bright light source, one Adjustable Analyzer (i.e., the polarizer), and a light sensor. How is the axis of the polarizer oriented when the glare is minimized?

2. Set two Adjustable Analyzers so that the polarizing axes are at 90°. Insert a third Adjustable Analyzer between the first two and collect data of the transmitted intensity as a function of the angle of the middle filter. Explain the shape of the graph using a vector model.